

Computational Neuroscience — Module 05: Action —

Study Questions

1. How does Active Inference's view of the motor cortex differ from the traditional "command" model?
2. What are proprioceptive predictions? How do spinal reflexes fulfill them?
3. Describe the basal ganglia direct ("Go") and indirect ("No-Go") pathways.
4. How does dopamine modulate the balance between Go and No-Go pathways?
5. Why does Parkinson's disease cause difficulty initiating movement?
6. Why does Huntington's disease cause involuntary movements?
7. What is the climbing fiber error signal in the cerebellum? What does it correct?
8. How do parallel fibers and Purkinje cells work together in motor learning?
9. Why does practicing a motor skill make it smoother? Explain using cerebellar circuits.
10. What are mirror neurons? Where are they found?
11. How does Active Inference explain mirror neurons without postulating a separate "mirror system"?
12. How do mirror neurons connect action to social cognition?
13. Describe the full cortico-basal ganglia-thalamo-cortical loop in sequence.
14. What role does the thalamus play in motor control?
15. How does cerebellar ataxia demonstrate the importance of forward models?
16. What is the difference between the basal ganglia's and the cerebellum's roles in action?
17. How might Tourette syndrome be understood as basal ganglia hyperactivity?
18. Why does movement feel effortless when it's well-practiced?
19. How does the motor system handle unexpected perturbations (e.g., tripping on a curb)?
20. Compare the computational neuroscience view of action with the cognitive science view (Unit 1, Module 05).